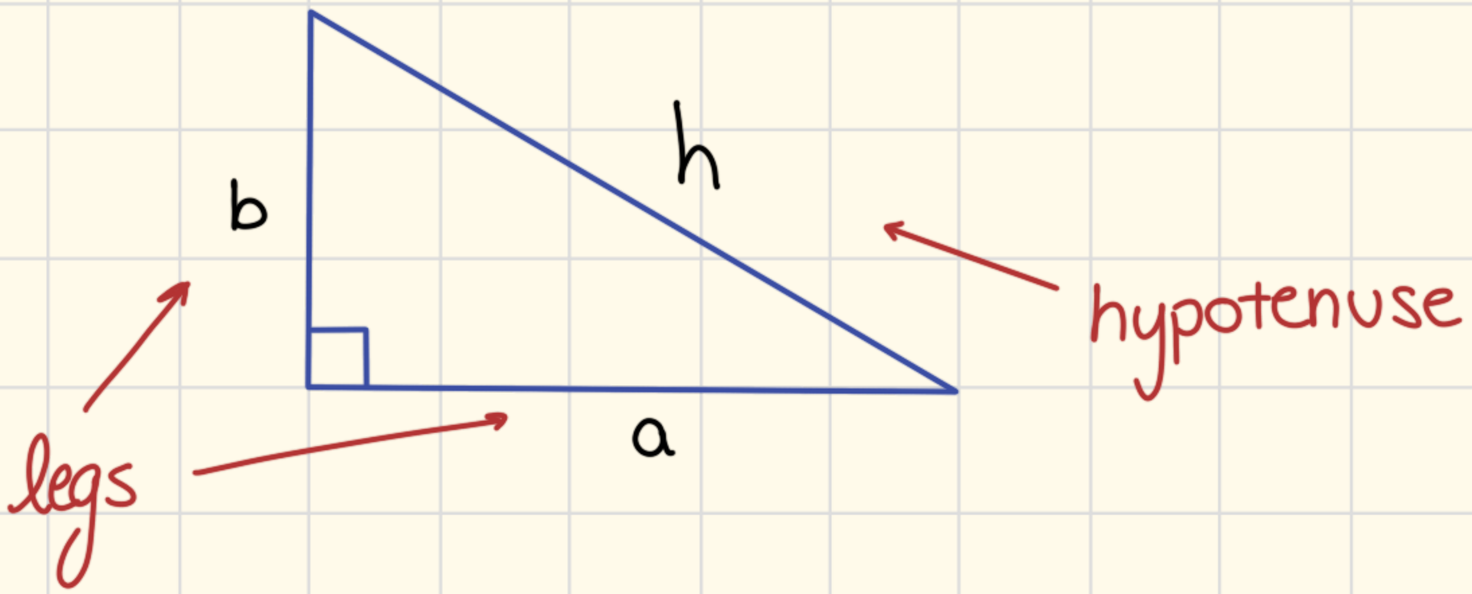
PYTHAGORUS THEOREM



$$h^2 = a^2 + b^2$$

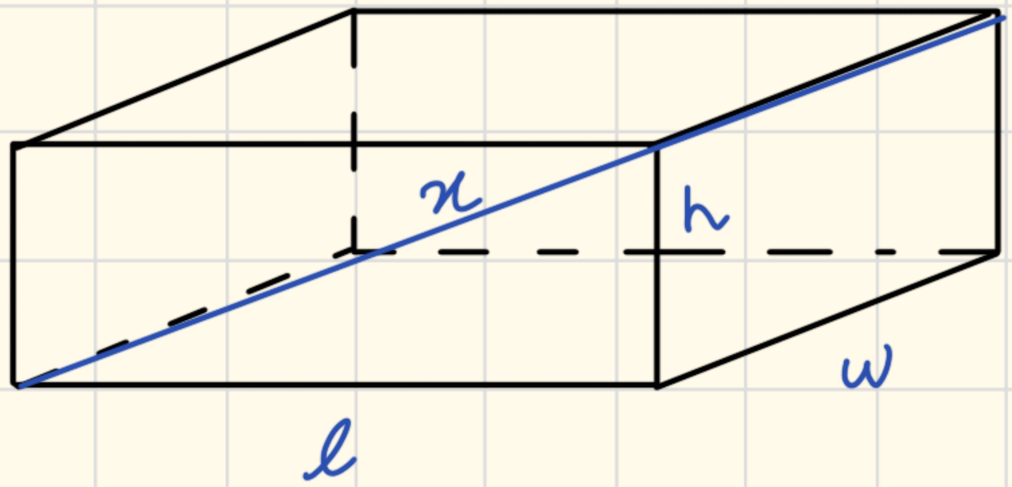
HYPOTENUSE

2 legs

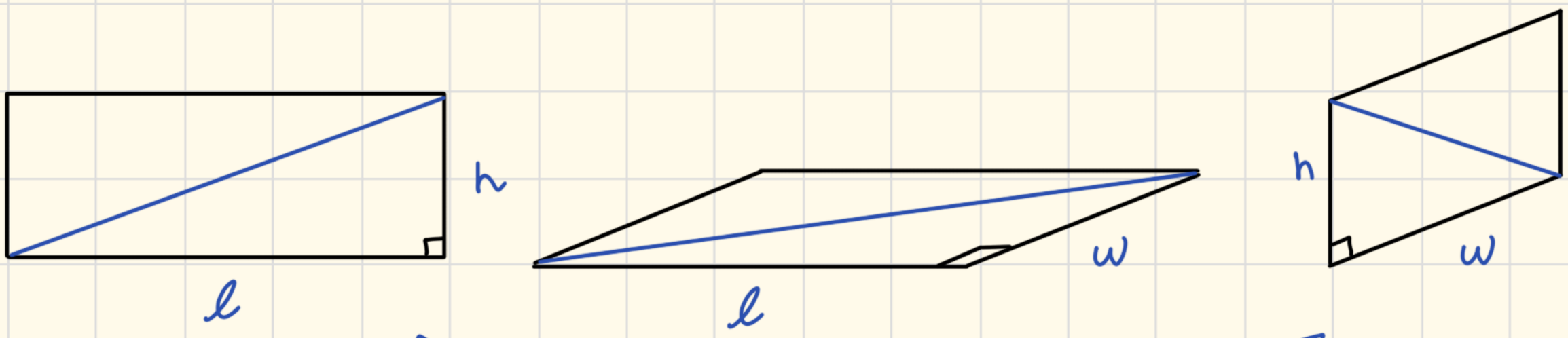
LEGS

a hypotenuse and leg

Pythagorus Theorem In 3D

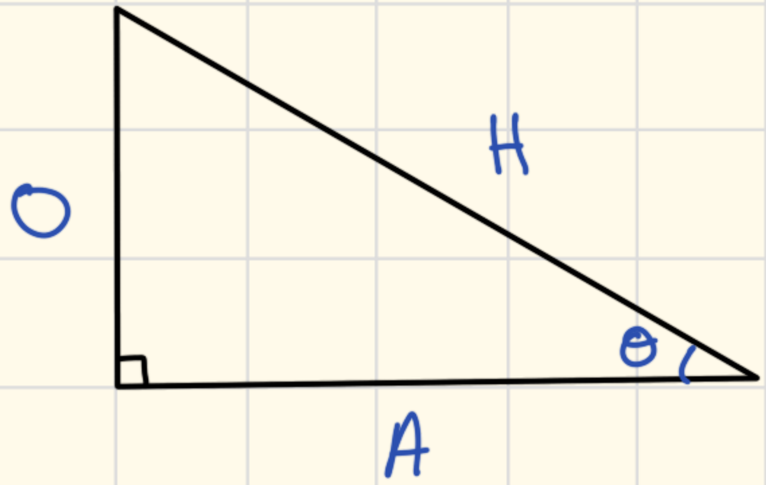


$$x^2 = \ell^2 + w^2 + h^2$$



Pythagorus In 2D

TRIGONOMETRY



⚠ REMEMBER

SOH - CAH - TOA

$$\sin\theta = \frac{O}{H} \qquad \cos\theta = \frac{A}{H} \qquad \tan\theta = \frac{O}{A}$$

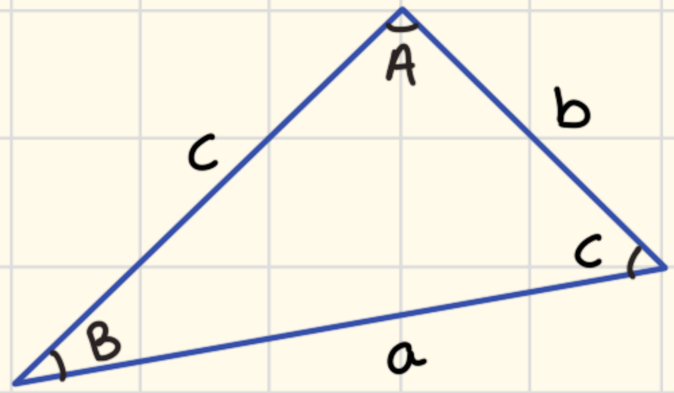
ANGLE

2 sides

SIDE

1 angle and 1 side

Sine Rule



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Sides
Angles

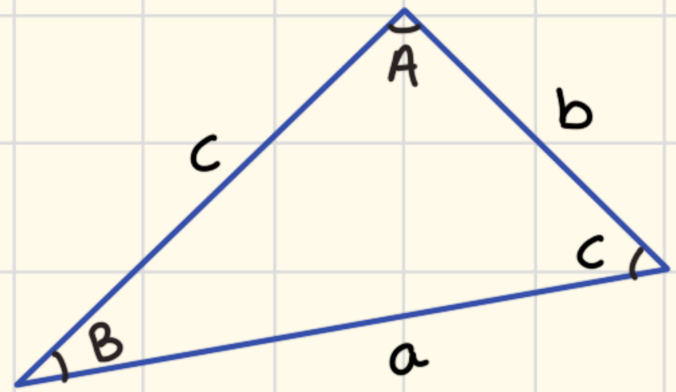
Angle

1 angle and 2 sides

Side

2 angles and 1 side

COSINE RULE



$$a^2 = b^2 + c^2 - 2bc\cos A$$

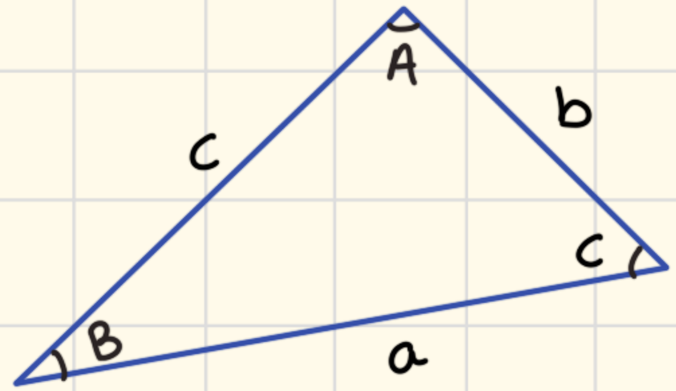
ANGLE

3 sides

SIDE

2 sides and opposite angle

Sine Area Of Triangle



$$\text{Area} = \frac{1}{2}\, ab \sin C$$

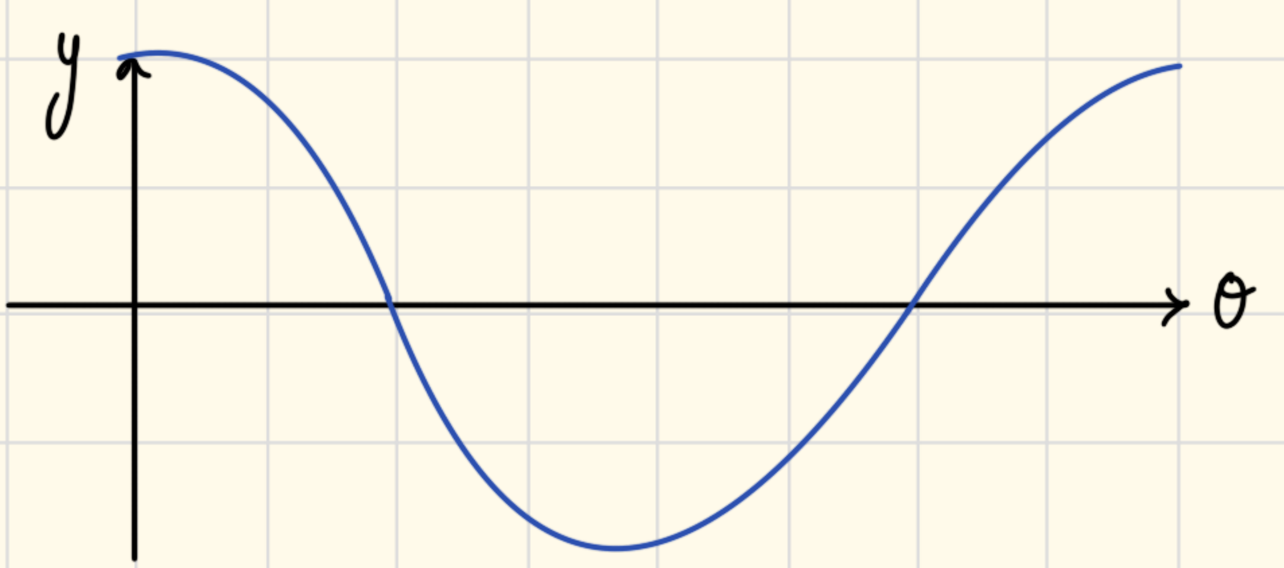
- To find area, you need

  2 sides & angle between them

TRIGONOMETRIC GRAPHS



$y = \sin\theta$

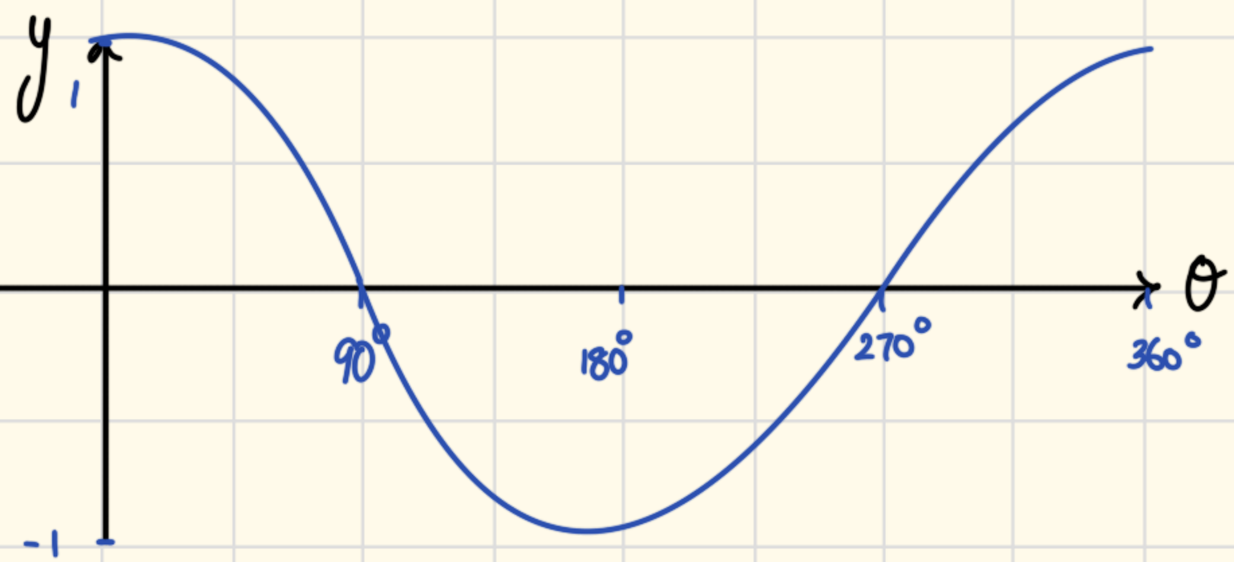


$y = \cos\theta$

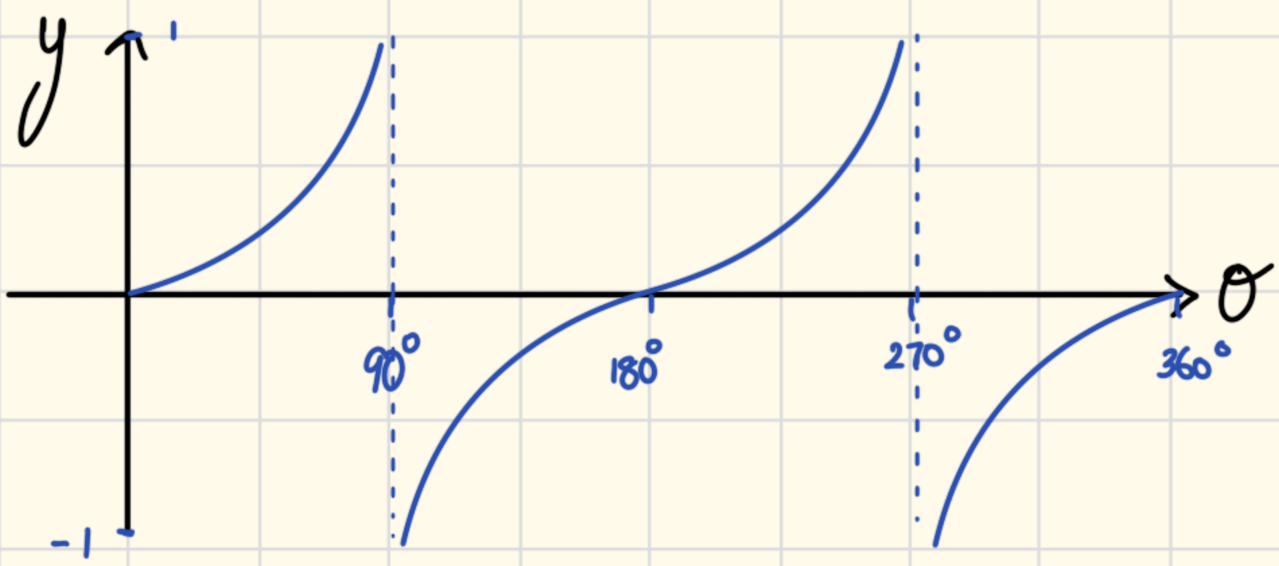
$y = \tan\theta$



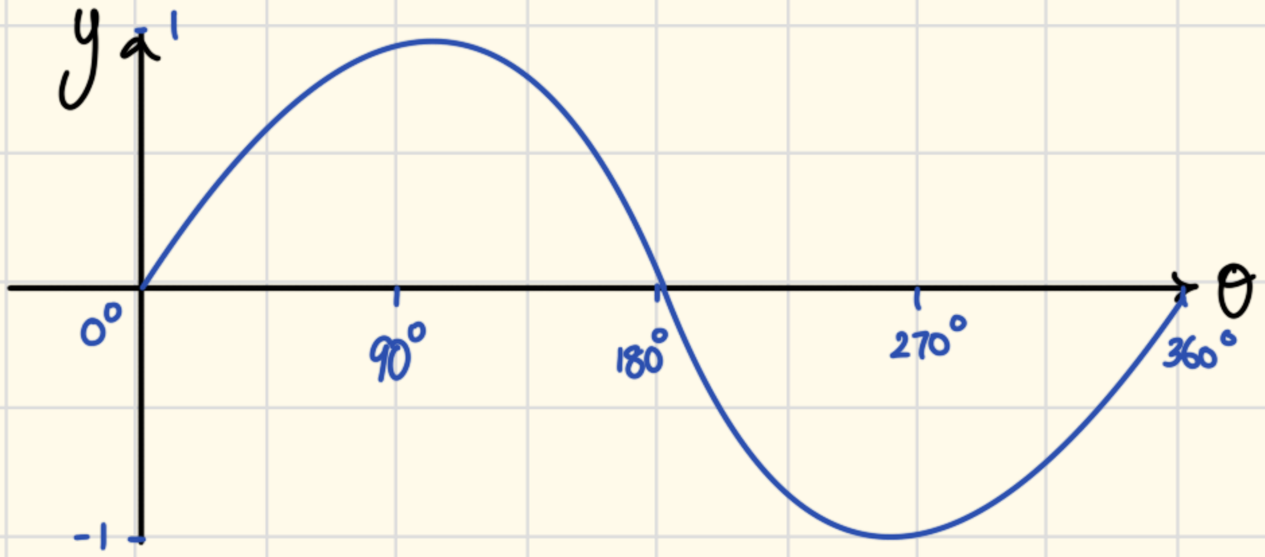
$y = \sin\theta$



$y = \cos\theta$

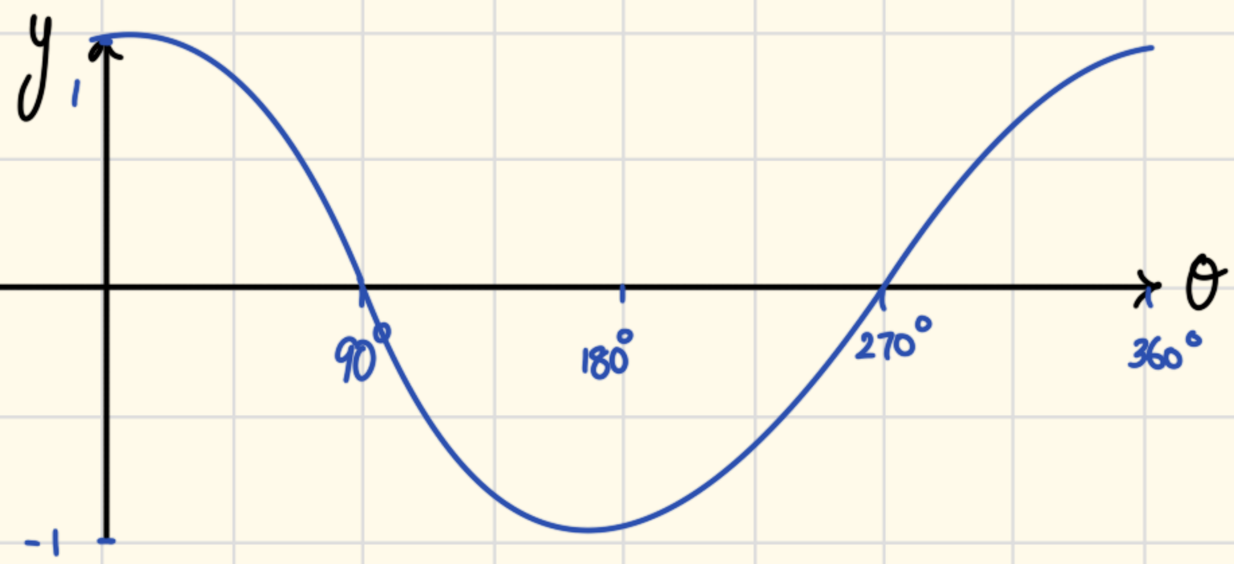


$y = \tan\theta$



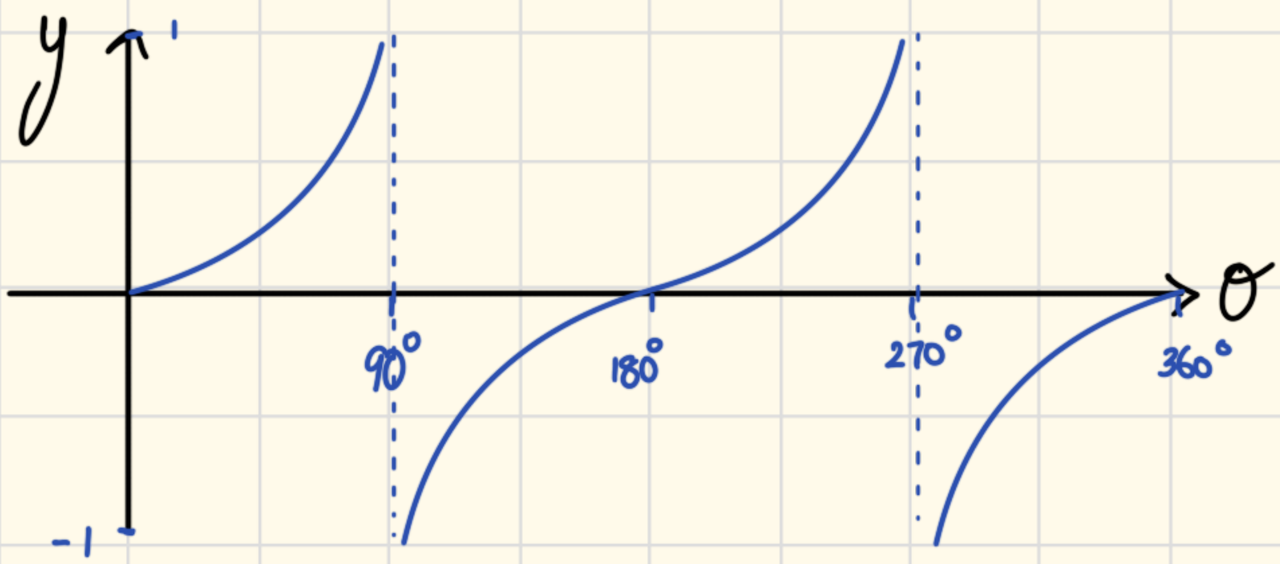
$y = \sin\theta$

θ	y
0°	0
90°	1
180°	0
270°	-1
360°	0



$y = \cos\theta$

θ	y
0°	1
90°	0
180°	-1
270°	0
360°	1



$y = \tan\theta$

θ	y
0°	0
90°	-
180°	0
270°	-
360°	0

Values Of Sine Function

$$\sin\theta = \frac{1}{2}$$

calculator gives 30°

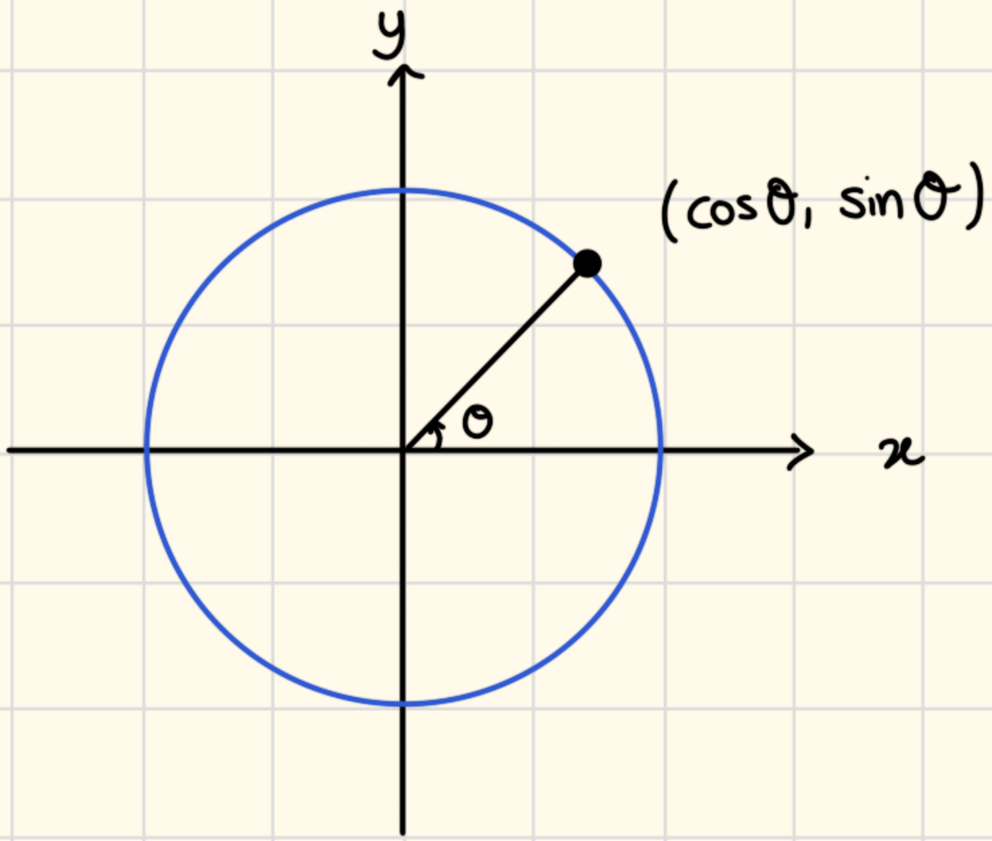
150° is an answer too!

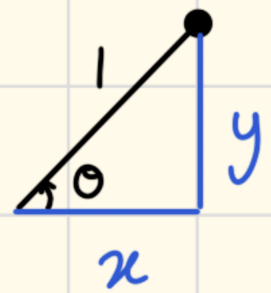


To get another answer for $\sin\theta = c$, take

$$180 - \theta^\circ$$

TRIGONOMETRIC FUNCTIONS IN CIRCLE

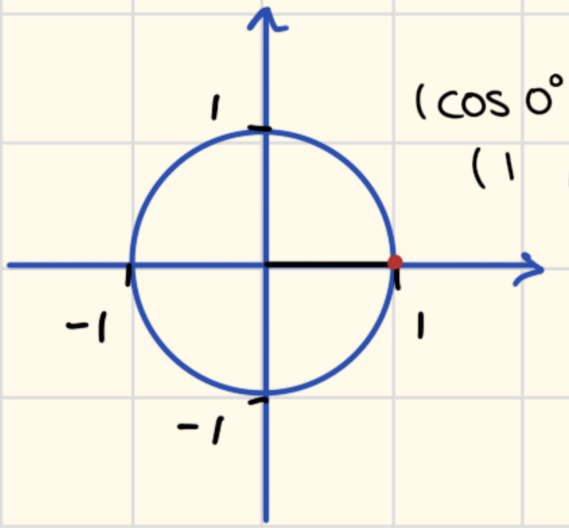




$\cos\theta = x$

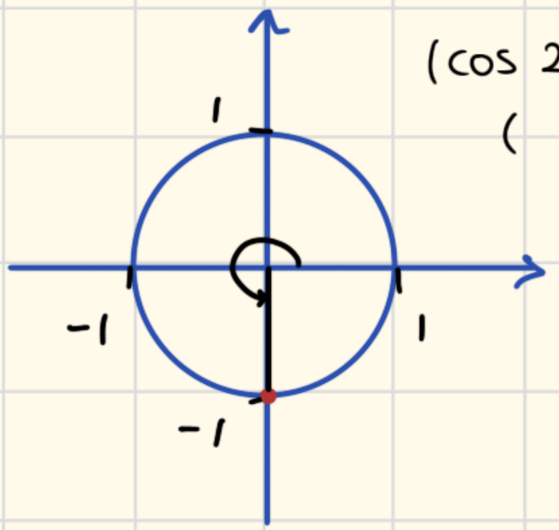
$\sin\theta = y$

Coordinates of points on circle $\Leftrightarrow$ Trigonometric functions!



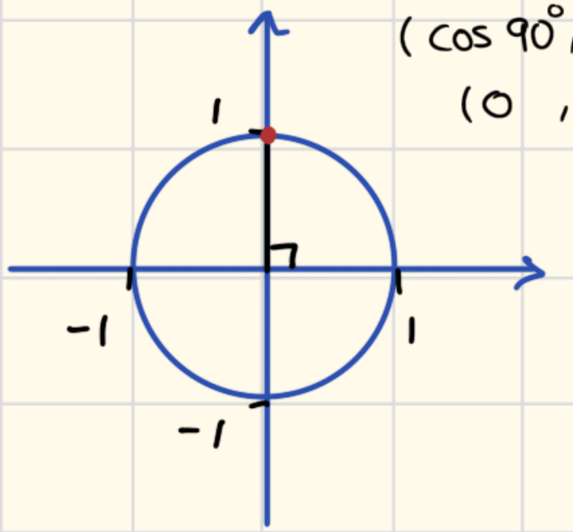
$(\cos 0^\circ, \sin 0^\circ)$

$(1, 0)$



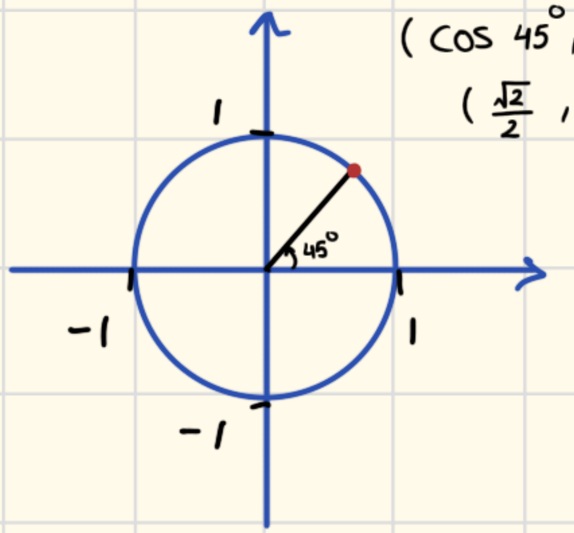
$(\cos 270^\circ, \sin 270^\circ)$

$(0, -1)$



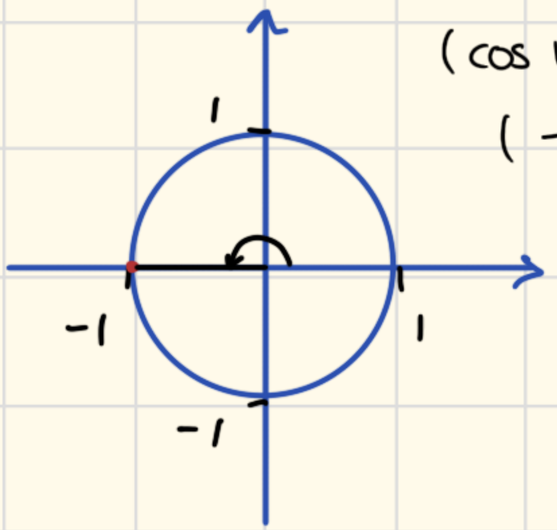
$(\cos 90^\circ, \sin 90^\circ)$

$(0, 1)$



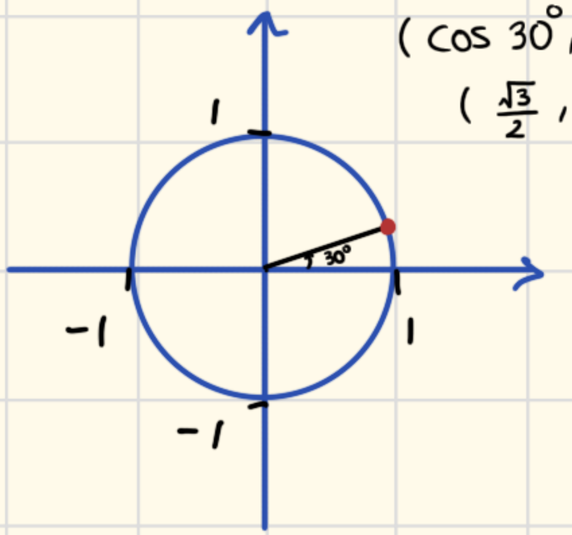
$(\cos 45^\circ, \sin 45^\circ)$

$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$



$(\cos 180^\circ, \sin 180^\circ)$

$(-1, 0)$



$(\cos 30^\circ, \sin 30^\circ)$

$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$